Do CO2 emissions predict legal condemnations among CAC 40 companies ?

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1. Abstract

This study examines whether CO2 emissions and dividend levels can predict the number of legal condemnations received by CAC 40 companies. Using a dataset covering the 40 companies of the French stock index, we conducted linear regression analyses to test three causal relationships suggested by a Directed Acyclic Graph (DAG): the effect of dividends on CO2 emissions, the effect of CO2 emissions on condemnations, and the direct effect of dividends on condemnations. Results show that CO2 emissions are a significant positive predictor of condemnations, while dividends show no significant direct effect once emissions are accounted for. These findings suggest that a company's environmental footprint is a stronger predictor of legal sanctions than its financial performance alone.

2. Introduction

Corporate legal liability has become an increasingly prominent topic in the context of environmental and social governance (ESG). Large publicly listed companies face growing scrutiny from regulators, NGOs, and the public regarding both their environmental impact and their financial practices. In France, the CAC 40 index comprises the country's largest and most influential corporations, making it an ideal sample for studying the relationship between economic activity, environmental damage, and legal accountability.

A company's CO_2 emissions may reflect the scale and nature of its industrial activities, which in turn expose it to legal risks. Meanwhile, the distribution of dividends serves as a proxy for financial performance and corporate governance choices.

This study investigates whether these two dimensions — environmental and financial — independently or jointly predict a company's number of legal condemnations. We formulate the following research question:

Do CO2 emissions and dividend levels predict the number of legal condemnations among CAC 40 companies ?

We structure our causal model using a Directed Acyclic Graph (DAG) with three nodes: Dividends, CO2 Emissions, and Condemnations. Dividends are hypothesized to influence both CO2 emissions (companies with higher profits may scale operations and thus emit more) and condemnations directly. CO2 emissions are hypothesized to influence condemnations, as heavier polluters are more likely to face environmental litigation (**see Figure 1**).

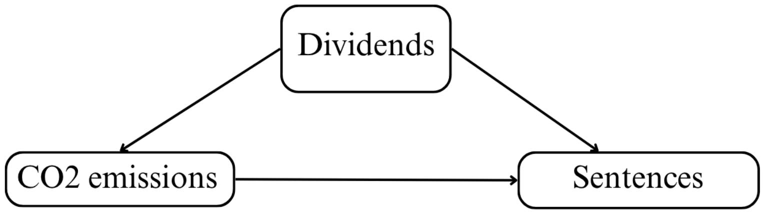

Figure 1 - Directed Acyclic Graph representing the causal links between our variables.

Prediction 1: *The higher the CO2 emissions, the higher the number of legal condemnations.*
Prediction 2: *The higher the dividends paid, the higher the number of legal condemnations.*

3. Data

The dataset was manually assembled from publicly available sources for all 40 companies composing the CAC 40 French stock index (N=40, **see Figure 2**). For each company, we collected : total revenues (CA), dividends distributed (div), total CO2 equivalent emissions (co2), and the total number of legal condemnations (nb_condamnations).

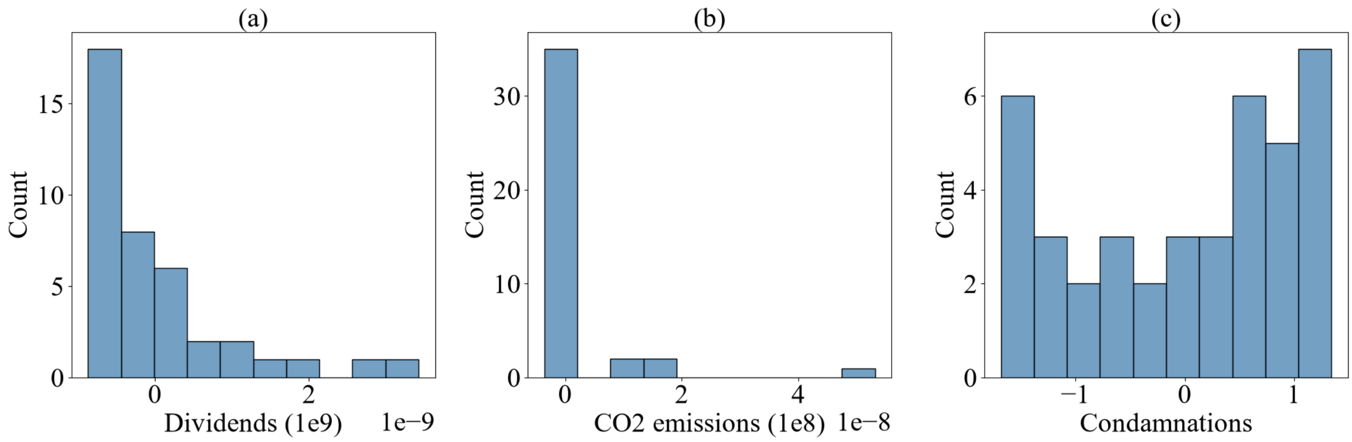

Figure 2 – Monovariate descriptions of the dataset

4. Methods

To test the prediction, we performed a multivariate regression test. We computed the following mixed linear model:

$$Sentences_i = \beta_0 + \beta_1 CO_2emissions_i + \beta_2 Dividends_i + \epsilon_i$$

And for each explanatory variable (CO_2 emissions, Dividends), βk is its regression coefficient, which measures its effect on our dependant variable, and whose significance is evaluated by its p-value.

5. Results

Companies that emit less CO2 are more likely to be penalized for environmental violations (β=-0.265, p=0.005, **see Figure 3**).This also seems to be the case for dividends, but we cannot rule out the possibility of a random correlation (β=-0.177, p=0.172, **see Figure 3**).

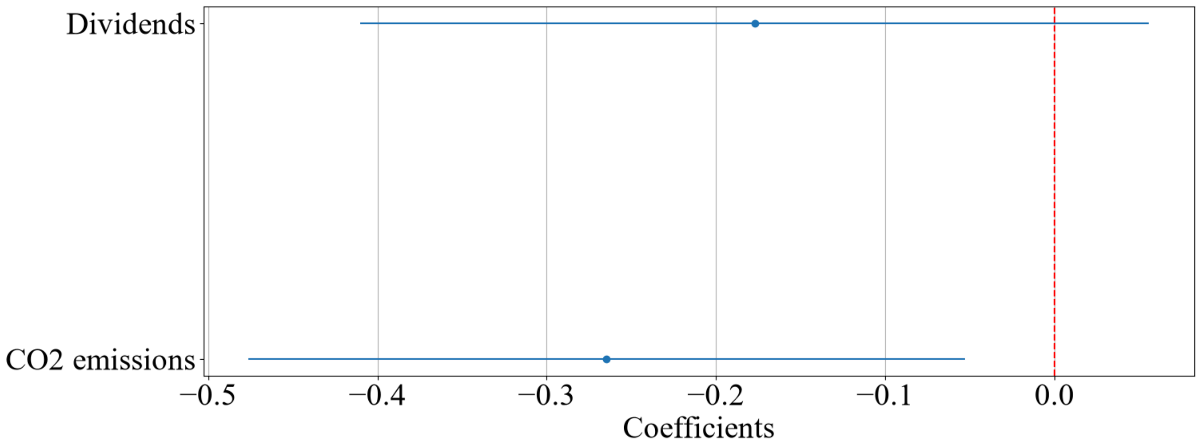

Figure 3 – Forest plot of the effects of dividends and CO2 emissions on condemnations.

6. Conclusion

This study finds that CO2 emissions are a meaningful predictor of legal condemnations among CAC 40 companies, while the direct effect of dividends appears to operate mainly through the mediation of emissions. These results suggest that a company's environmental impact is more central to its legal exposure than its financial profile per se. The weak explanatory power of the models, however, calls for caution: the number of legal condemnations is a complex outcome shaped by institutional, sectoral, and strategic factors that a simple linear framework cannot fully capture. Future work should incorporate sector fixed effects, longitudinal data, and more granular measures of environmental and governance performance.